ExpeL: LLM Agents Are Experiential Learners

Andrew Zhao, Daniel Huang, Quentin Xu, Matthieu Lin, Yong-Jin Liu, Gao Huang · Tsinghua University · AAAI version · arXiv 2308.10144v3

HarnessRSI 72 ★★★

evidence: retained

Paper

Code

1 What changes, what is kept

1 Gather (training tasks)
ReAct actor; on failure Reflexion retries, up to Z = 3. Every trajectory, success or fail, goes to the experience pool. Signal: environment success / fail

2 Extract insights (gpt-4-0613)
Reads a fail/success pair of the same task, or L successes, and edits ONE insight list. Starts empty.

3 Use at test time
Prompt = task + full insight list + top-k most task-similar successful trajectories (Faiss). gpt-3.5-turbo-0613, ONE attempt, no retries

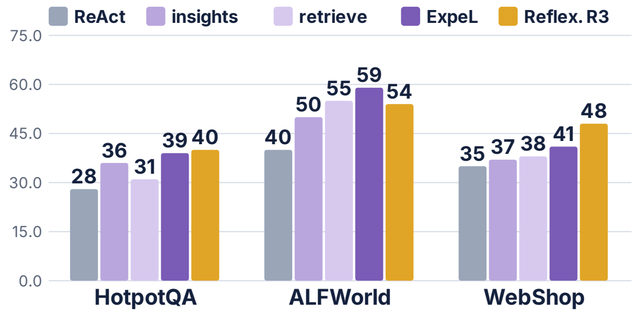
Keep / reject rule for each insight

ADD	new, count 2	UPVOTE	count +1
EDIT	rewrite, +1	DOWNVOTE	count -1

count reaches 0 ⇒ insight removed

weights never change; kept = insight list + trajectory pool
text-only actor frozen at gpt-3.5-turbo-0613

2 Headline: 3 benchmarks, one attempt



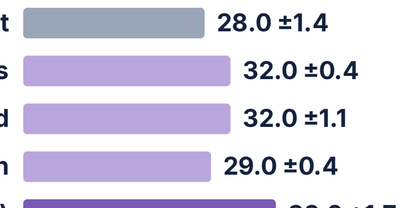
Success rate %, gpt-3.5 actor (Fig. 5), mean over folds. Gold bar retries on the test tasks; ExpeL does not. Table 3: HotpotQA 28.0 ± 1.4 to 39.0 ± 1.7, ALFWorld 40.0 ± 0.3 to 59.0 ± 0.3. Test sets 100 / 134 / 100 tasks.

4 Where the gain is absent or thin

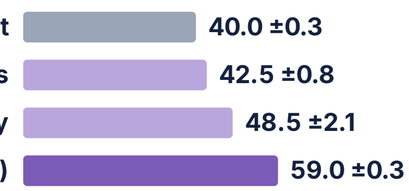
- **HotpotQA:** Reflexion R3 reaches 40, ExpeL 39 (single attempt). **WebShop:** ExpeL 41 vs Reflexion R1 43, R3 48
- gpt-4 insight layer adds +8 / +4 / +3 over retrieve-only (31 / 55 / 38). With gpt-3.5 as extractor HotpotQA is 32.0 vs ReAct 28.0
- ALFWorld vs ReAct by type (Table 5): large on 4 of 6, **clean +13**, **cool −4** (67 vs 71). Per-type counts not given
- **Cost:** tokens per trajectory, ExpeL vs ReAct: 4310.06 vs 1319.75 (HotpotQA), 2856.7 vs 2051.49, 3291.31 vs 2575.41. Gathering + gpt-4 cost, leakage check: not reported

3 Controls that decide if the gain is real

Insight source · HotpotQA SR % (Table 3)



Retrieval rule · ALFWorld SR % (Table 3)



Experience · HotpotQA (Fig. 6)

few-shot only 27 · ReAct-only experience 35 · full 39 (ReAct 28)

Transfer HotpotQA to FEVER (Table 1)	SR %
Act / ReAct	58 / 63
transfer, no task demos	65 ± 1.7
transfer, with demos	**70 ± 0.7**

Not run: ReAct / Reflexion never get the training experience (retrieve-only does) or the gpt-4 extractor; no matched-token row; temp 0, no repeated runs; one transfer pair, insights rewritten by gpt-4.

Takeaway

Keeping an insight list and similar successes from earlier tasks lifts a frozen gpt-3.5 agent over ReAct, and both parts matter. But on HotpotQA most of the gain needs a gpt-4 extractor the baselines do not get, Reflexion with retries matches or beats ExpeL on two of three benchmarks, and build cost is not reported.

Our read · HarnessRSI

Score 72 · ★★★ · evidence level: retained

Reports: held-out tasks, fold mean ± SE, ablations, one transfer.
Does not report: matched-compute baselines, build cost, leakage check.